rate.

A generic quantum state can be thought of as a superposition of a set of basis states with definite fixed energy. (Just like a generic state of a spinning electron can be thought of as a superposition of an electron that is definitely spin-up and one that is definitely spin-down.) The Hamiltonian tells us what the actual energy is for each possible definite-energy state. Given that list of possible energies, we can ask whether any particular way of dividing the wave function into subsystems implies that those subsystems interact "locally." In fact, for a random list of energies, there won't be any way of dividing the wave function into local subsystems, but for the right kind of Hamiltonian, there will be exactly one such way. Demanding that physics look local tells us how to decompose our quantum system into a collection of degrees of freedom.

In other words, we don't need to start with a set of fundamental building blocks of reality, then stick them together to make the world. We can start with the world, and ask if there is a way to think about it as a collection of fundamental building blocks. With the right kind of Hamiltonian, there will be, and all of our data and experience of the world suggests that we do have the right kind of Hamiltonian. It's easy to imagine possible worlds where the laws of physics weren't local at all. But it's hard to imagine what life would be like in such a world, or even whether life would be possible; the locality of physical interactions helps bring order to the universe.

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We can begin to see how space itself emerges from the wave function. When we say that there's a unique way of dividing up our system into degrees of freedom that interact locally with their neighbors, all we really mean is that each degree of freedom interacts with only a small number of other degrees of freedom. The notions of "local" and "nearest" aren't imposed from the start—they pop out from the fact that these interactions are very special. The way to think about it isn't "degrees of freedom interact only when they are nearby," but rather "we *define* two degrees of freedom to be 'nearby' when they directly interact with each other, and 'far away' when they don't." A long list of abstract degrees of freedom has been knit together into a network, in which each degree of freedom is connected to a